

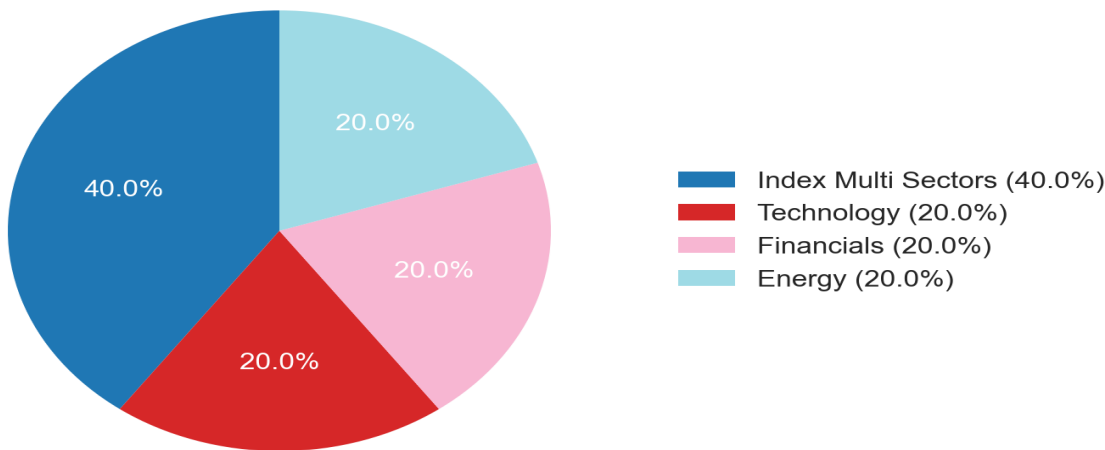
Portfolio rotation report

Date of extraction	2025-12-07 22:09
Data window	2020-01-01 → 2024-10-01
Frequency	Monthly — 58 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	QQQ, SPY, XLE, XLF, XLK
Indicators	Short term: 5 months volatility, 9 months return, 12 months momentum Long term: 12 months volatility, 18 months return, 36 months momentum

1. Portfolio Composition

- XLK (Technology Select Sector SPDR Fund — Technology)
- XLE (Energy Select Sector SPDR Fund — Energy)
- XLF (Financial Select Sector SPDR Fund — Financials)
- SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
- QQQ (Invesco QQQ Trust — Index Multi Sectors)

Sector Allocation



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

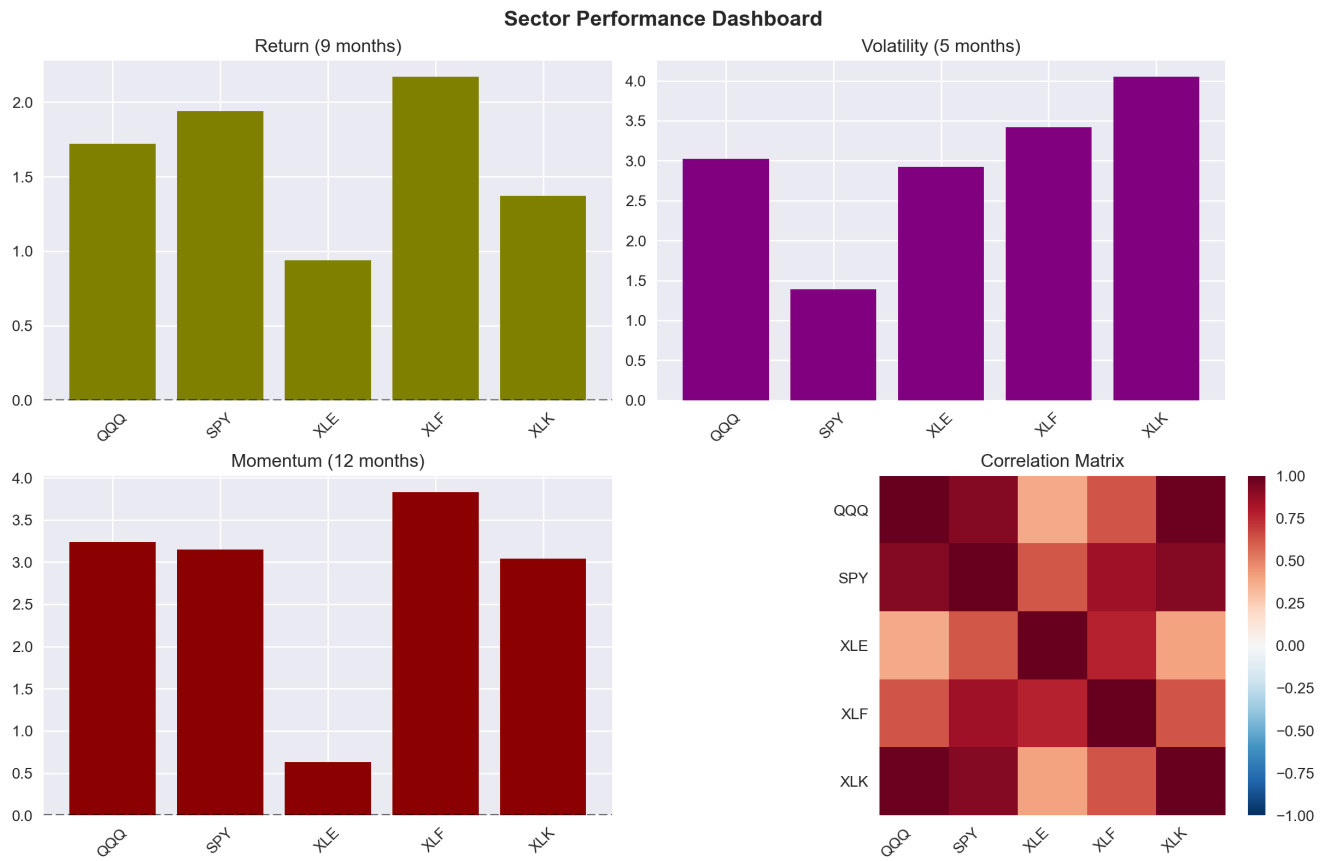
Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Market-neutral beta exposure: close alignment with index dynamics.
- Volatility profile skewed to dynamic/speculative names.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations
100.0% of portfolio

Pair	Correlation
QQQ / XLK	0.98
QQQ / SPY	0.92
SPY / XLK	0.92
SPY / XLF	0.85
XLE / XLF	0.77

Overall, average cross-sector correlation was 0.71, ranging from 0.38 to 0.98. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion remains meaningful, indicating both tightly and weakly linked sectors. Strongest co-movement between QQQ and XLK (0.98); weakest between QQQ and XLE (0.38). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
QQQ	1.72	3.02	0.568	3.24	strongly coherent	subdued	liquid	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, low volatility — calm market tone, with ample liquidity.
SPY	1.94	1.39	1.396	3.15	strongly coherent	subdued	liquid	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, low volatility — calm market tone, with ample liquidity.
XLE	0.94	2.92	0.323	0.63	mildly coherent	subdued	liquid	positive performance with solid momentum, signal remains coherent, but with limited conviction, low volatility — calm market tone, with ample liquidity.
XLF	2.17	3.42	0.634	3.83	mildly coherent	subdued	thin	positive performance with solid momentum, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
XLK	1.37	4.05	0.338	3.04	strongly coherent	normal	thin	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
XLF	2.17
SPY	1.94
QQQ	1.72
XLK	1.37
XLE	0.94

Top by Volatility

Ticker	Volatility(%)
XLK	4.05
XLF	3.42
QQQ	3.02
XLE	2.92
SPY	1.39

Top by Momentum

Ticker	Momentum(%)
XLF	3.83
QQQ	3.24
SPY	3.15
XLK	3.04
XLE	0.63

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
QQQ	-0.50	4.19	0.624	0.75	1.18	53.05 M	stable performance, momentum surging, volatility compressing, amplifying market trends, under thin liquidity regime.
SPY	0.15	3.86	0.648	0.45	1.00	72.91 M	stable performance, momentum surging, volatility collapsing, amplifying market trends, under thin liquidity regime.
XLE	0.55	0.31	0.481	0.86	0.80	28.61 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLF	0.24	6.25	0.528	0.91	0.84	36.84 M	stable performance, momentum surging, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLK	-0.73	2.66	0.831	0.83	1.33	16.33 M	stable performance, momentum surging, volatility compressing, amplifying market trends, under thin liquidity regime.

■ High-impact observations

For **XLK**: *Technical rebound* — improving momentum while returns are still limited. (avg ann. return 1.4%, momentum 3.0%, vol 4.0%)

■ Moderate-impact observations

For **QQQ, SPY, XLF**: *Technical rebound* — improving momentum while returns are still limited. (avg ann. return 1.9%, momentum 3.4%, vol 2.6%)

■ Light-impact observations

For **XLE**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.9%, momentum 0.6%, vol 2.9%)